- The acceleration due to gravity is denoted by g m s^{-2}. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

9 The diagram shows a toy caterpillar consisting of a head and three body sections each connected by a light inextensible ribbon. The head has a mass of 120 g and the body sections each have a mass of 90 g.

The toy is pulled on level ground using a horizontal string attached to the head. The tension in the string is 12 N. There are resistances to motion of 2.5 N for the head and each section of the body.



(a) (i) State the equation of motion for the toy caterpillar modelled as a single particle. **[2]**

(ii) Calculate the acceleration of the toy caterpillar. **[1]**

(b) Draw a diagram showing all the forces acting on the head of the toy caterpillar. **[3]**

(c) Calculate the tension in the ribbon that joins the head to the body. **[2]**